



# Chiroptera: a Bio-Inspired Leisure Sailboat

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BIOS-4740-A/ Fall 2025/ Instructor:  
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# Opportunities & Existing Solutions

# Design Statement

*Our project translates biological folding and structural strategies into a high-performance leisure sailboat. By combining a whale-inspired hull for stability and spaciousness with a bioinspired, efficient sail system, we create a craft that deploys rapidly, adapts to wind conditions, and optimizes onboard space. This synthesis of biomimicry and elegant design elevates both performance and user experience.*



# Existing Problems

## Aerodynamic and Performance Limitations

- Traditional sails operate efficiently only within a narrow wind range, losing lift under shifting or light winds.
- Fabric deformation and heeling forces reduce energy efficiency and directional stability.

## Structural and Material Constraints

- Conventional rigs lack flexibility; rigid masts and soft sails cannot dynamically adjust to variable loads.
- UV, salt, and flex fatigue shorten lifespan; folding or reefing mechanisms add bulk and complexity.

## Human, Control, and Integration Challenges

- Sail trimming remains manual and labor-intensive, requiring constant human input for performance.
- Poor aerodynamic-hydrodynamic coordination causes drag and instability during maneuvering.



# Prior Art: Existing Designs/ Solutions



## Improved Sail Shape

- Aerodynamic and performance limitation improvements
- Modern sail profiles use advanced airfoil geometry and tension control to maintain optimal lift across varying wind angles, reducing drag and heeling while improving overall aerodynamic efficiency.



## Michelin: Inflatable Sails

- Structural and materials constraints exploration
- The inflatable wing-sail uses pressurized air chambers instead of rigid spars, creating lightweight, self-supporting structures that resist deformation and fold compactly.



## Junk Rig: Sectional Sails

- Human, control, integration challenges, and durability
- Its multiple battened panels enable easy reefing and self-balancing control, reducing manual workload and improving sail shape stability across different wind conditions.





### **Balogh Sail Designs (BSD) BATWING sail**

Handglide-inspired airfoils, Bat-like battens  
- basic structure, not shape or form.

- Small, Compact sail mainly used for kayaking



### **Richard Dryden's Transition Rig**

- Designed for use on kayaks and windsurfers, with concepts that can scale and integrate onto medium and large vessels.
- The design draws inspiration from bird wing skeletal structures, bat wing membrane folding, and a set of custom bio-inspired joints.

**Inspiration**  
**n**

# Design Theme & Focused Object



## Locomotion

Designs that directly effect movement or interact/ contact direct movement.

X



## Sailboat

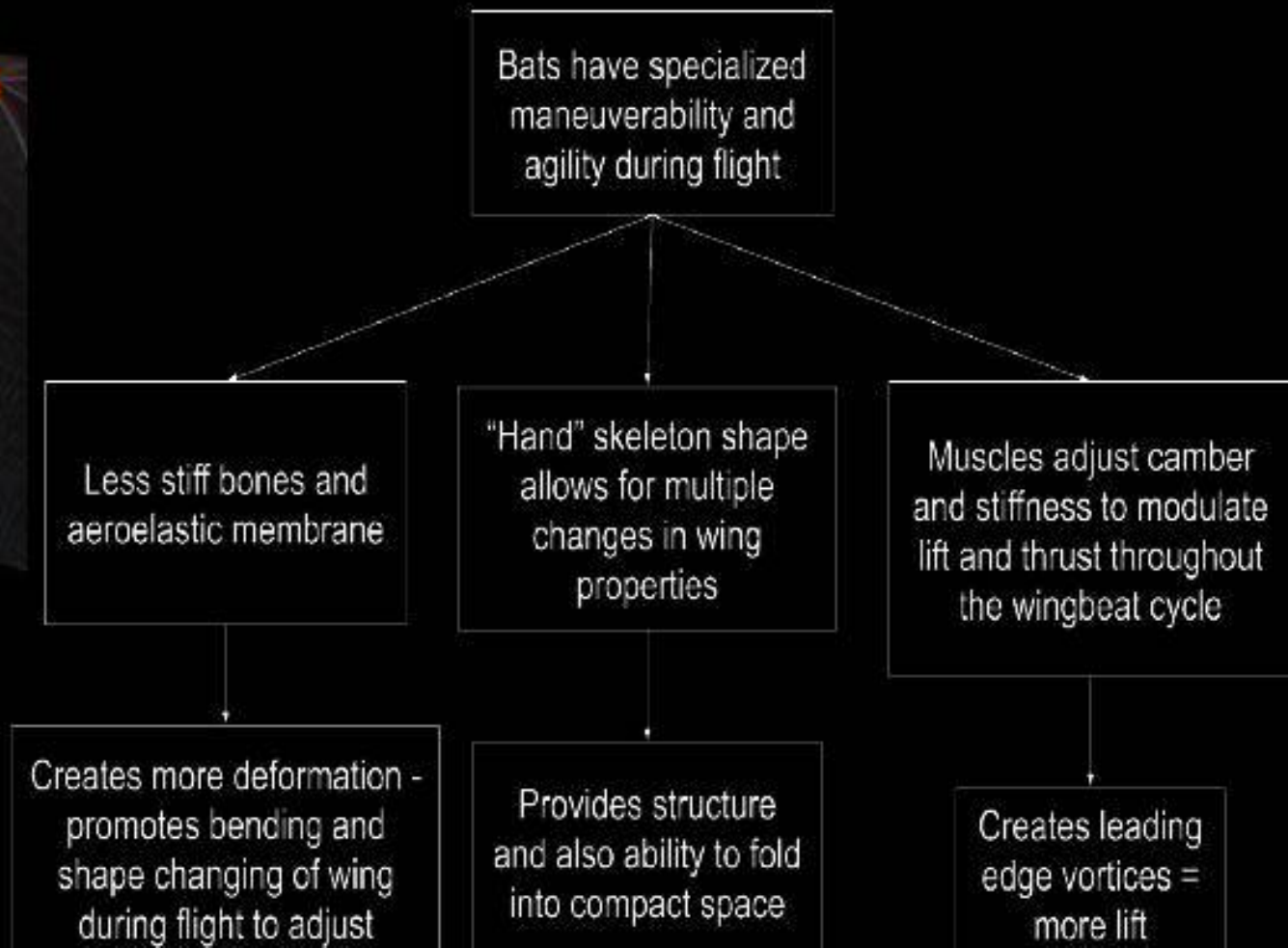
A boat that is propelled mainly by sails using the power of the wind, though many also carry small engines or oars for maneuvering.



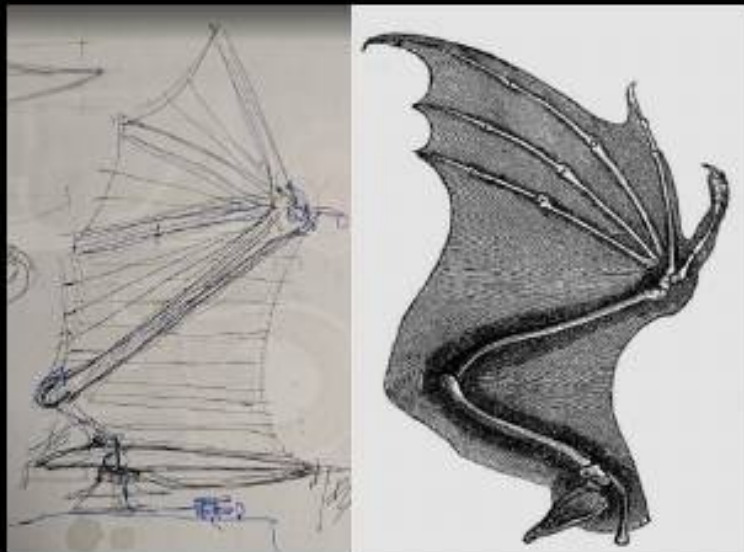
## Choosing Organisms



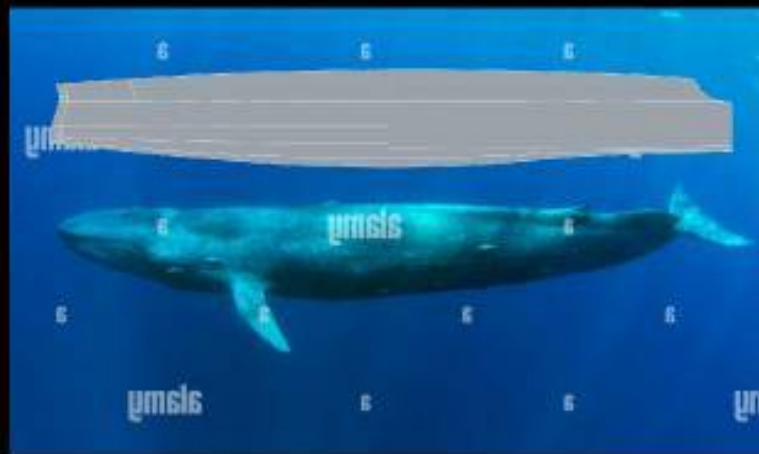
## Selected Organisms



## Biological Inspiratio Summary



Bat wings combine lightweight skeletal ribs with tensioned membranes, allowing controlled deformation and high maneuverability. This anatomy informs the sail's rib-membrane system, enabling variable camber, smooth folding, and real-time aerodynamic adaptation.



Whales embody streamlined mass and gentle curvature that balance hydrodynamic efficiency with volumetric spaciousness. Their broad midsection and tapered ends inspired the hull's smooth flow profile and expanded interior volume for a stable, luxury-oriented form.

# Concept Translation

source form/  
shape/ placement  
based on bat  
wing, duck feet



Headsai

| Mainsai

|

Mast

Boo

m



folding structure  
(origami) based  
on bat wing and  
plant inspiration  
sources, i.e.

Hul

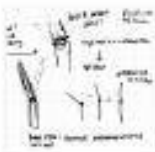
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Optimize hull form  
based on whale  
body/form

**Translated Components and  
Superlatives**

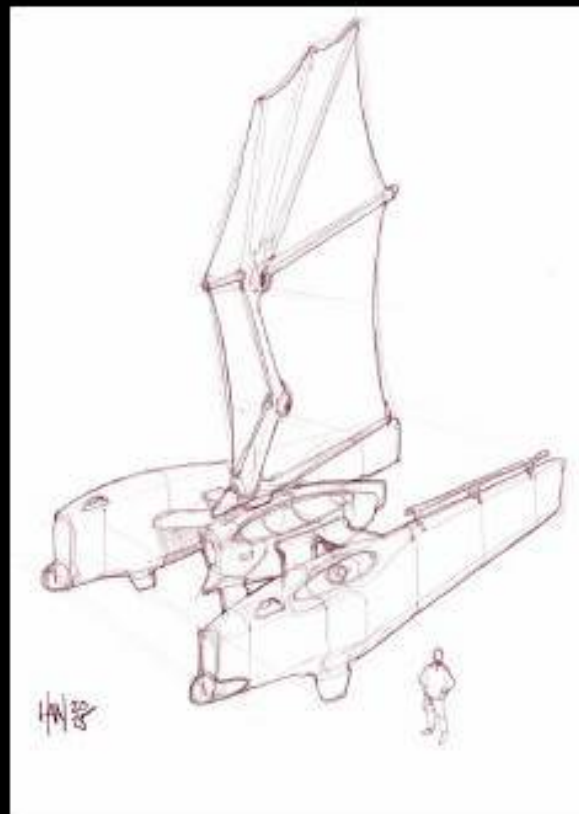
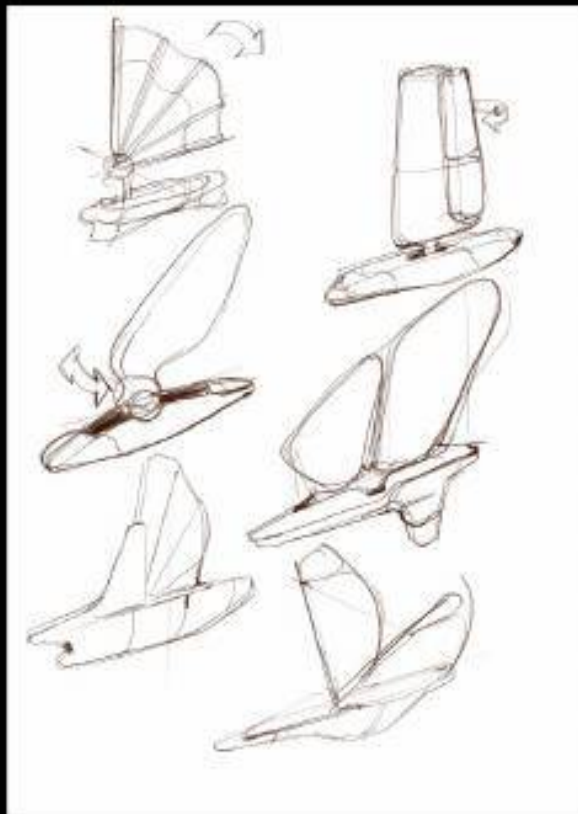


|                  | Design 1                                                                          | Design 2                                                                          | Design 3                                                                           | Design 4                                                                            | Design 5                                                                            | Design 6                                                                            |
|------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Lifting          |  |  |  |  |                                                                                     |                                                                                     |
| Adaptability     |  |  |  |  |  |                                                                                     |
| Sail Foldability |  |  |  |  |  |  |
| Space Efficiency |  |  |  |  |  |  |

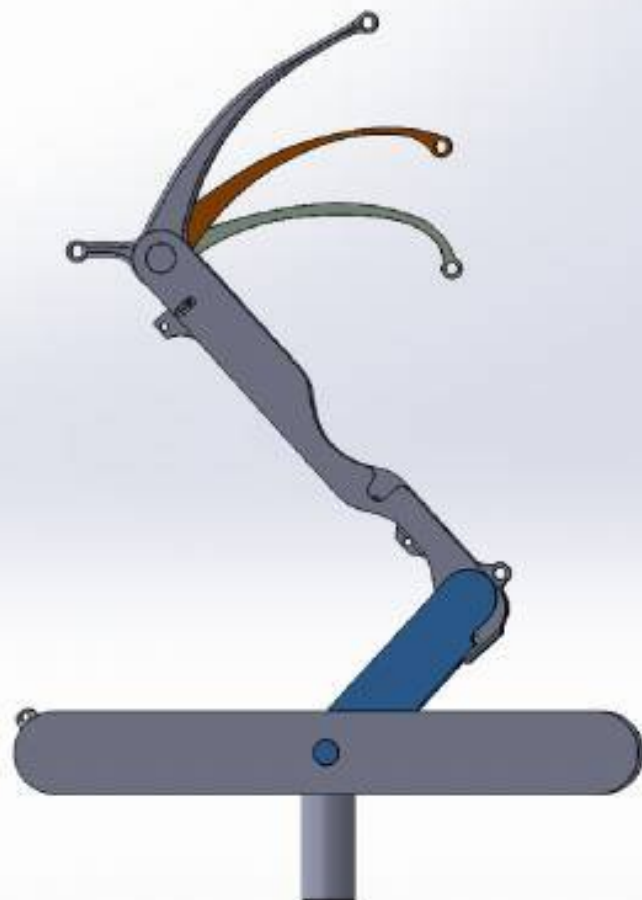
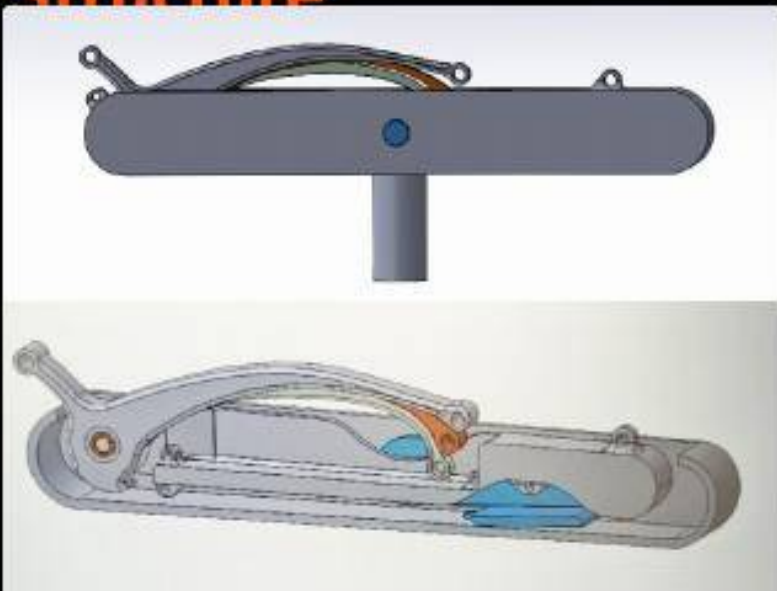
Evaluation  
Matrix

# **Design & Prototyping**

# Initial Concept Sketch



## Sail Design: Structure



The mast folds on 3 joints, similar to the three joints of the bat wing. The mechanism will use a rope to extend the mast up and down via a pulley system connect to a winch. Improvements on this model are in development.

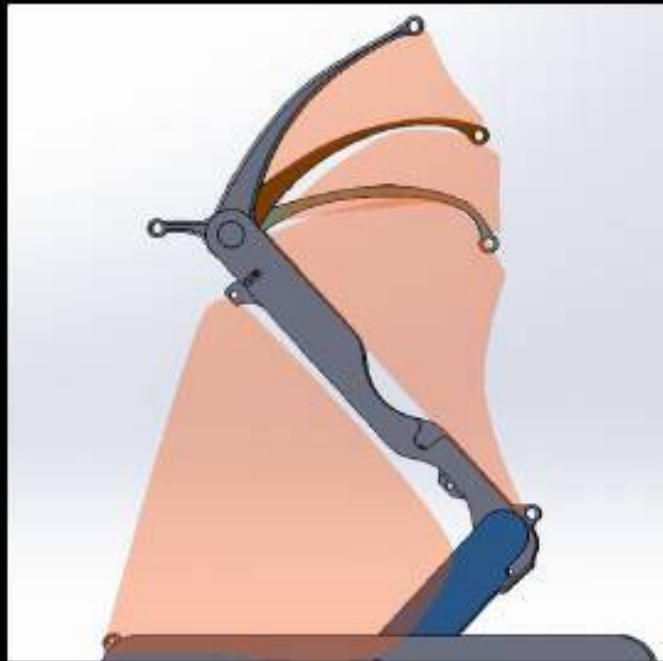
## Sail Design: Folding Mechanism



Mechanism can be triggered via a single wire pull action, enhancing user experience and efficiency



## Sail Design Membrane



Bat-inspired ribs support flexible membranes, distributing stress evenly and allowing adaptive sail curvature for smoother aerodynamic performance. Wind tests will be done on the model to simulate its aerodynamics.

● = radius, ulna (fused), claw, and first two fingers (fused)

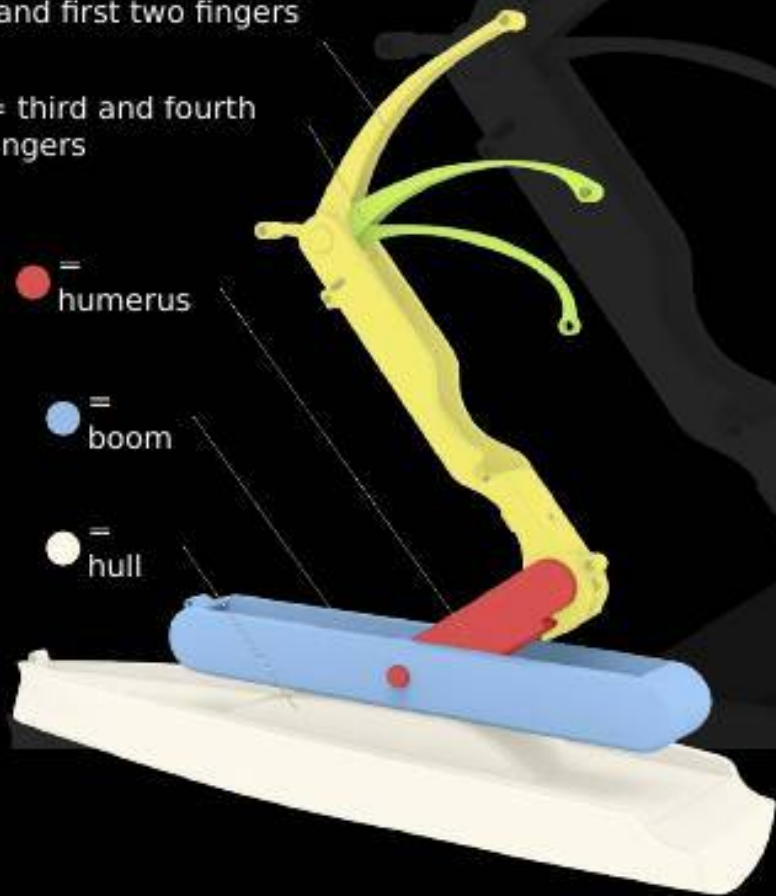
● = third and fourth fingers

● = humerus

● = boom

● = hull

## Rough Structural Breakdown

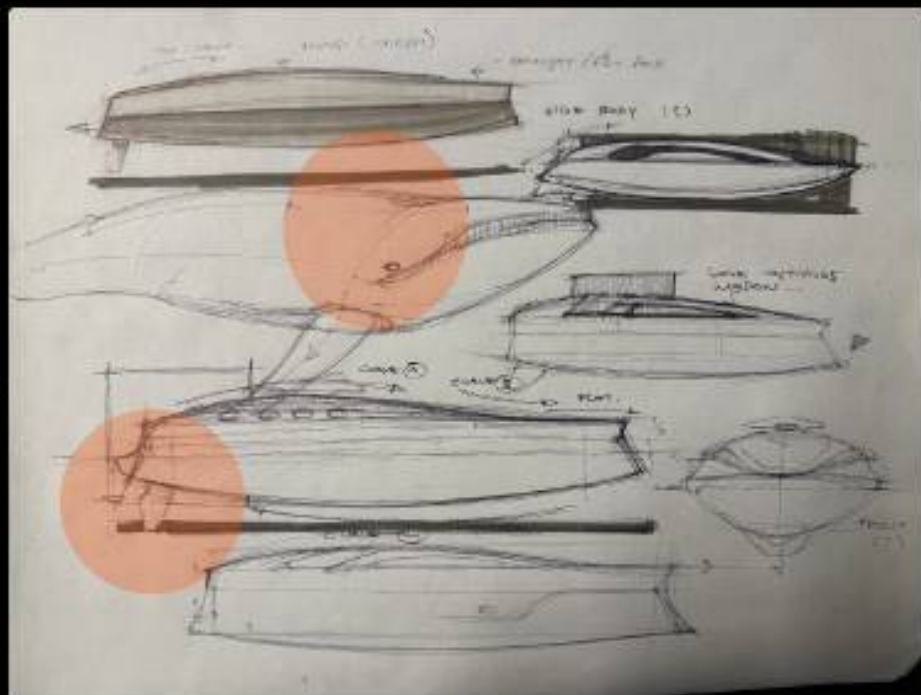


# Hull Design



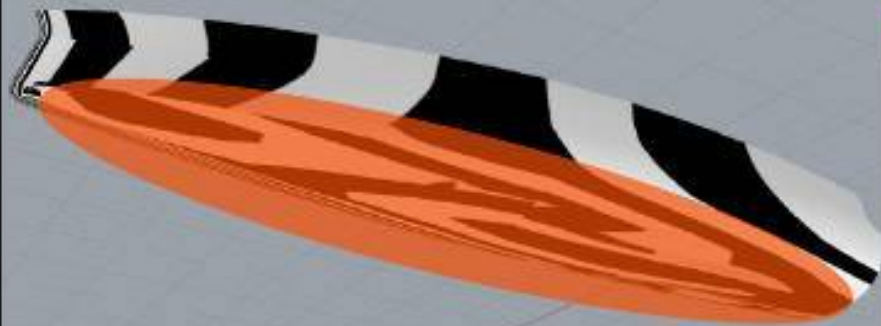
**Whale-inspired hull, focusing on spaciousness and stability.**

Profile echoes the fluid silhouette of a whale, while the front view retains a sailboat's keel form, widening gracefully near the deck to enhance interior volume and balance.



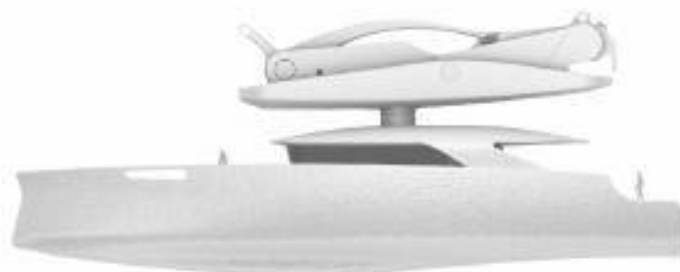


Design Iterations of Hull Design Modeled in Rhino 3D



Zebra Stripe Analysis demonstrates surface discontinuity in modeled solid

## Hull Refinement

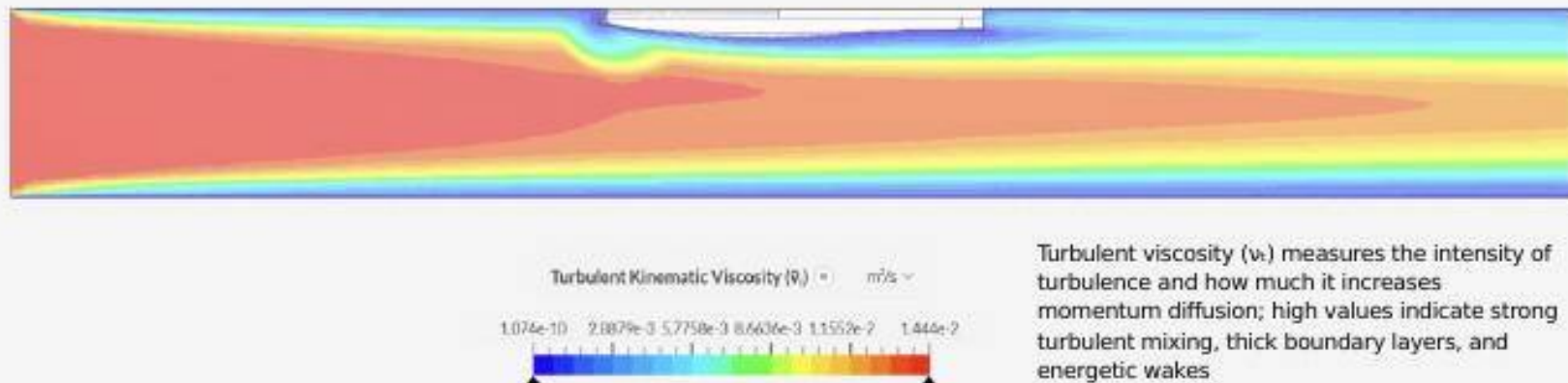


Finalized hull design w/ improved surface quality and better hydrodynamics



**Hull  
Refinement**

# Hull Assessment via CFD



- Wake turbulence correctly forms behind the hull.
- Low turbulence near bow and under hull indicates stable boundary layer behavior.
- Upstream region shows abnormally high turbulence, which should not occur in calm inlet flow.\*



# Prototyping



# Material

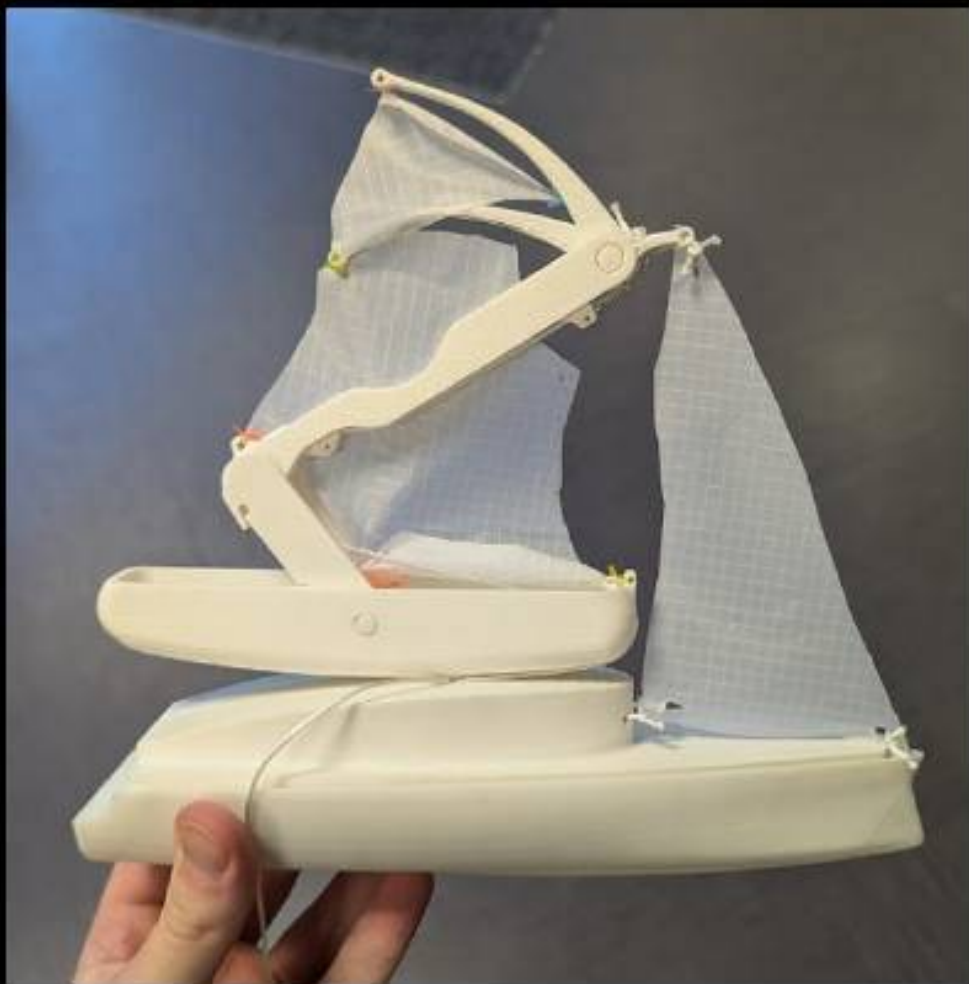


Sail: Rip-Stop Nylon (nylon 6,6 with polyurethane coating) thin and lightweight yet waterproof and impermeable to air

Boom/Mast: PLA (polylactic acid); lightweight plastic; most commonly used additive manufacturing, relatively high stiffness and decent tensile strength

Hull: UV-curable acrylate-based photopolymer resin; denser and more sturdy than PLA with no visible print lines, helps lower boat's center of mass

**Testin  
g**



# Sail Test Procedure

- Place Prototype in pool
- Fan used to simulate wind
- Our design vs normal mast/sail
  - Test distance traveled
  - time for X distance





# Chiropter a

## Purpose/Environment

- Large sailboats
- Luxury sailing, daysailing

## Inspiration

- Bat wing structure and membrane
- Bio-inspired joints
- Whale-inspired hull

## Attributes

Large sail

Both sail and mast collapse completely

Greater space efficiency (whole boat)

Adjustable geometry and fortification

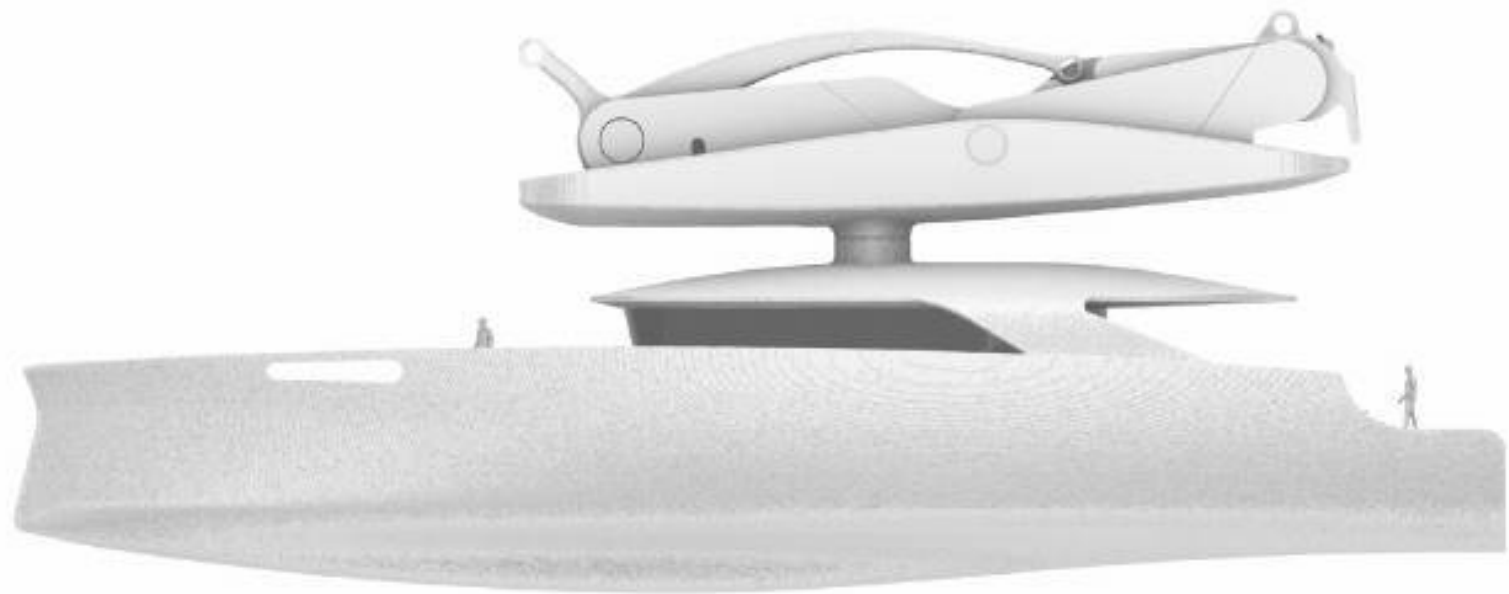
Better maneuverability











# Source

## S

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# THANK YOU

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